



Morgan Stanley Global Macro Forum

Chipflation – Implications of a Memory Crisis

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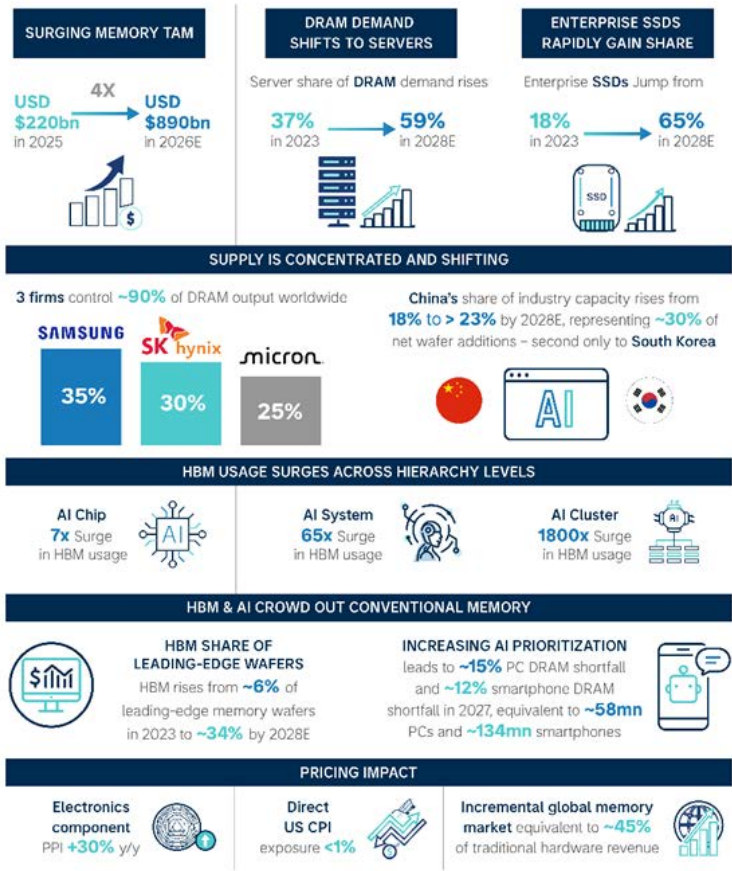
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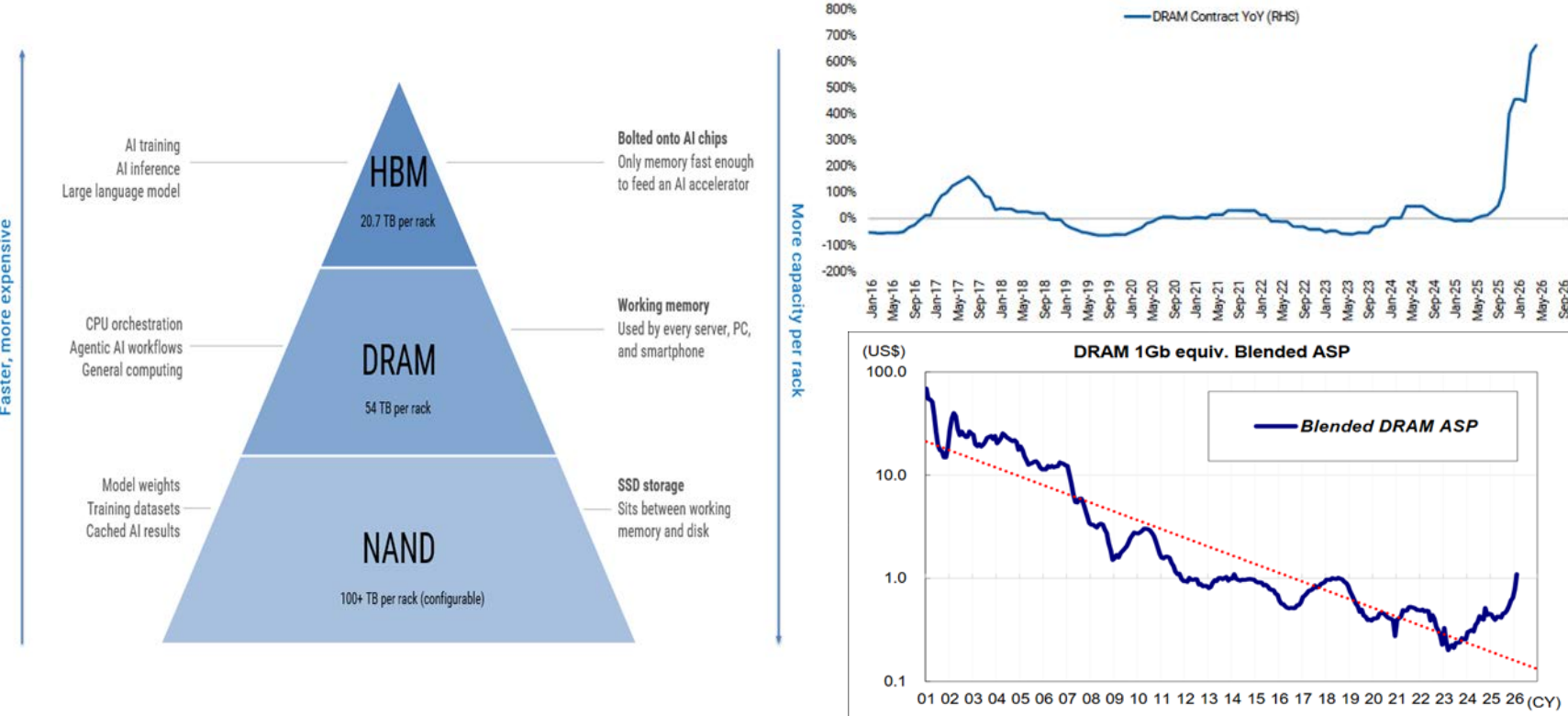
Chipflation – The Story in Numbers



Source: Morgan Stanley Research

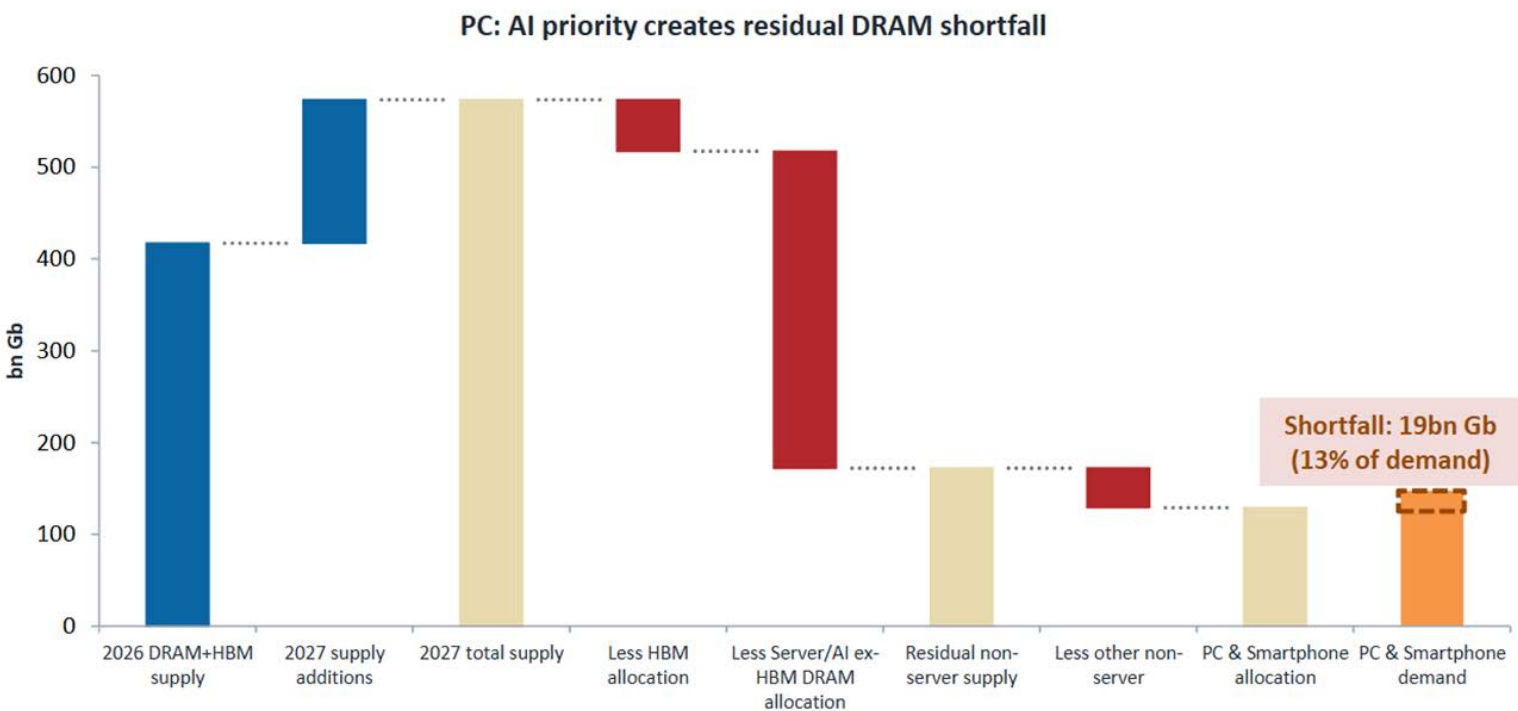
AI Is Turning Memory Into a Structural Bottleneck

AI servers are becoming memory systems



Source: WSTS, TrendForce, Morgan Stanley Research

AI Prioritization Turns 2027 Supply Growth Into a Consumer Memory Shortfall



Source: Morgan Stanley Research

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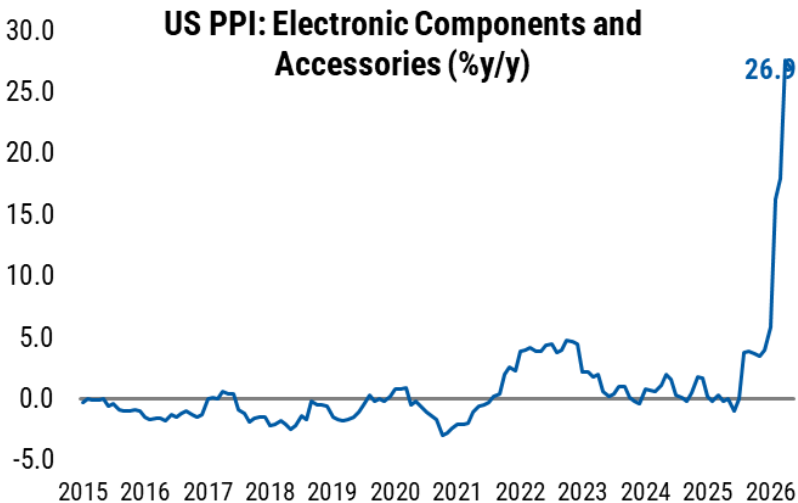
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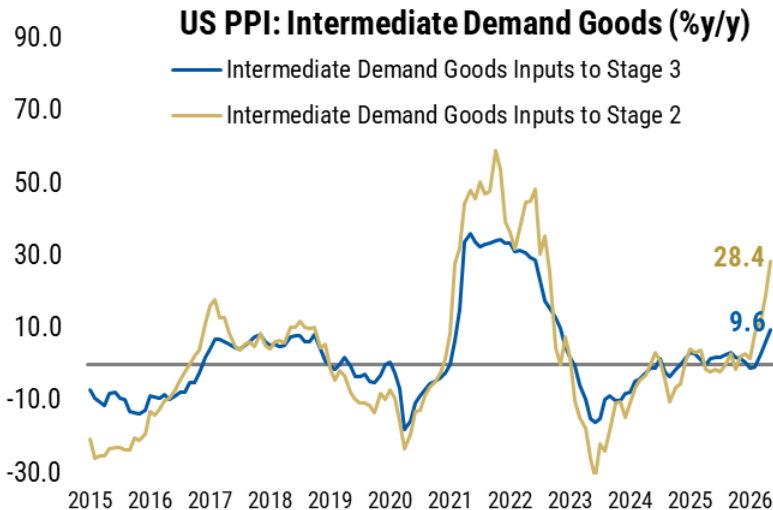
Chipflation Is Boosting Production Costs

The effects of memory costs on inflation are most visible in PPI electronics and related equipment. Higher costs could push inflation but also squeeze margins, with implications to growth.

Sharp rise in electronics...



... boosting upstream input costs



CPI Impact Will Likely Be More Micro Than Macro

We think that the impact on CPI will be more micro than macro – visible in individual categories, but with a modest effect on the overall index, in the order of ~10bp to headline inflation in 2026.

CPI impact of higher memory cost

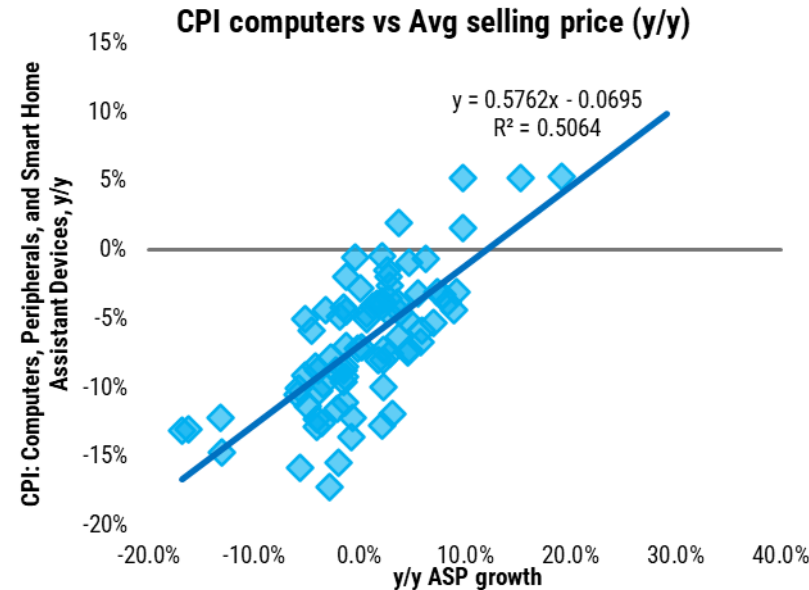
CPI component	Memory costs effect on CPI in 2026 (pp)	CPI weight
PCs	15	0.30%
Smartphones	15	0.20%
TVs	10	0.10%
Cars	0	3.80%
Major household appliances	5	0.06%
Game consoles	125	0.01%
Impact on headline CPI: PCs and smartphones only	0.08	
Total impact on headline CPI	0.10	

Source: BLS, Morgan Stanley Research forecasts; Note: Smartphone and game console weights are estimates.

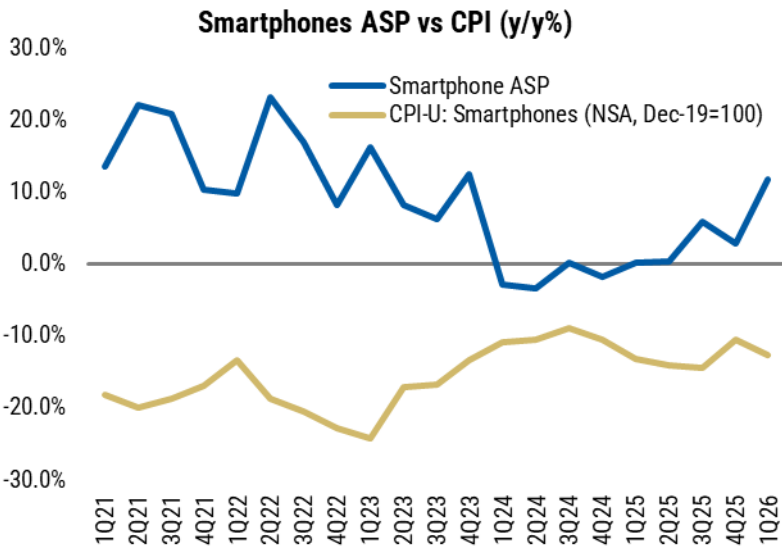
The Relationship Between ASPs and CPI Is Not One-to-One

CPI indexes for electronics are quality-adjusted, whereas ASP series are not. And this adjustment seems to make CPI series smoother compared to ASPs.

For PCs, a 30% rise in ASP is consistent with ~10% in CPI...



... and the link is unclear for smartphones



Source: BLS, Morgan Stanley Research

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Key Takeaways

- **Memory price move is a structural reset:** The era of memory getting reliably cheaper (Moore's Law) seems to have ended with memory prices having risen more than sixfold over the past year. The memory market is on track to grow from ~\$220bn in 2025 to ~\$890bn in 2026, according to TrendForce forecasts. That incremental ~\$600bn is larger than the entire standalone TAM for smartphones, PCs or servers. This is not a normal cyclical upswing, but a structural reset in the price behavior of a foundational input.
- **Is this a supply or demand problem?:** The price surge exists because AI has created a sudden, price-inelastic leap in memory demand; without it there is no cost shock. Inelastic demand meeting inflexible supply is what turns a cyclical upswing into a structural reset.
- **Chipflation Passthrough and Sector Impact:** This memory shortage is reshaping PC and smartphone markets as suppliers prioritize higher-value AI/server demand over lower-margin consumer applications. We now expect surging memory costs to lower 2026 PC shipments by 10%+ and smartphone shipments by 13% versus 2025, with memory and SSD price inflation driving higher device prices and demand increasingly skewing toward premium devices.
- **Macro implications:** Chip Inflation is already visible in producer price inflation (PPI) segments tied to computing and electronics. The effect on CPI is not as strong, perhaps suggesting difficult margin decisions being made by consumer-facing producers.
- **US/China policy options:** Even if the US and China make policy choices aimed at alleviating price pressure, actually doing so will take years. Moreover, our expectation is that US policy choices are skewed toward a more, not less, restrictive direction.

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Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

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